CNC-Lathe — Standard Operating Procedure

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> **Audience:** Maintenance Tech · Line Manager · Quality Worker
> **Applies to:** Line-A, station 1, CNC-Lathe

1. Purpose

The CNC-Lathe is the entry point of Line-A (Precision Machining). It accepts raw bar stock and produces a rough turned shaft profile for the downstream CNC-Mill. As the first station, an outage on this lathe starves every downstream station on the line within minutes.

2. Normal operating envelope

- Ideal cycle time: **40 s** (`idealCycleTimeSeconds`)
- Max acceptable cycle time: **48 s** (`maxCycleTimeSeconds`)
- Expected reject rate: **≤ 2 %**
- Fault probability per cycle: **0.5 %**
- Input buffer capacity: **5 parts**
- Telemetry interval: **10 s**

3. Idle reasons

`idle_reason`	Meaning on the CNC-Lathe	First check
`Starved`	Raw bar stock feed empty. As station 1 this means the upstream feeder ran out — refill bar stock magazine.	Verify bar feeder + check the Production Schedule for an unscheduled gap.
`Blocked`	Downstream CNC-Mill cannot accept the next part — output buffer full.	Check CNC-Mill status; resolve its `Fault` or `Maintenance` first.

4. Faults & corrective actions

4.1 `Chuck-Jam`

- **Symptoms in telemetry:** `machine_status` flips to `Fault`, `fault_type = "Chuck-Jam"`. Actual cycle time on the prior cycle often exceeded 48 s as the chuck struggled. CNC-Mill (station 2) reports `idle_reason = "Starved"` within one telemetry interval.
- **Likely root cause:** A chip or fragment of bar stock has lodged between the chuck jaws, or the chuck pressure regulator has drifted.
- **Corrective action:**
  1. Apply LOTO per `Safety_Lockout_Tagout_Procedure.pdf`.
  2. Open the chuck, remove the bar stock remnant and any chips.
  3. Inspect each jaw for galling; replace the jaw set if scoring is visible.
  4. Verify chuck pressure at the regulator (target: 60 psi ± 5).
  5. Bump-cycle three test parts before releasing to production.
- **Restart criteria:** three consecutive parts complete within `idealCycleTimeSeconds` × 1.10 and no reject is raised.

4.2 `Bearing-Wear`

- **Symptoms in telemetry:** `fault_type = "Bearing-Wear"`; if not yet faulted, watch for rising actual cycle time (still under 48 s) and a slow uptick in reject rate above 2 %.
- **Likely root cause:** Spindle bearing has accumulated radial play beyond the 0.005 mm tolerance.
- **Corrective action:**
  1. LOTO. Open the spindle housing.
  2. Measure radial play with a dial indicator. If > 0.005 mm, replace the bearing pack as a matched set.
  3. Re-pack with the lathe OEM’s grease specification only.
  4. Run the 60-second warm-up cycle before resuming production.
- **Restart criteria:** spindle runout ≤ 0.005 mm verified post-repair.

4.3 `Tool-Break`

- **Symptoms in telemetry:** `fault_type = "Tool-Break"`. The reject before the fault will almost always be flagged in the Quality dashboard.
- **Likely root cause:** Tool insert exceeded service life or struck an inclusion in the bar stock.
- **Corrective action:**
  1. LOTO. Index the turret to the broken tool position.
  2. Replace the insert (or full holder if the holder shank is damaged).
  3. Re-set tool length offset against the master probe.
  4. Inspect the next 5 parts on the CMM (station 5) at 100 %.
- **Restart criteria:** post-repair tool length within 0.01 mm of master, no scrap on the next 5 parts at CMM-Inspection.

5. Preventive maintenance schedule

- **Daily:** check coolant level, blow off chip tray, verify bar feeder supply.
- **Weekly:** inspect chuck jaws, lube turret indexer.
- **Monthly:** spindle vibration analysis; replace coolant filter.
- **Quarterly:** spindle bearing endplay measurement; recalibrate tool length probe.

Missed PMs escalate per `Maintenance_Workflow_Overview.pdf`.

6. References

- Maintenance_Workflow_Overview.pdf
- Safety_Lockout_Tagout_Procedure.pdf
- Quality_Reject_Disposition_Policy.pdf
- Production_Lines_Reference.pdf (Line-A part flow)
- Line-A_02_CNC-Mill_SOP.pdf (downstream peer)
- Line-A_05_CMM-Inspection_Calibration.pdf (quality gate)